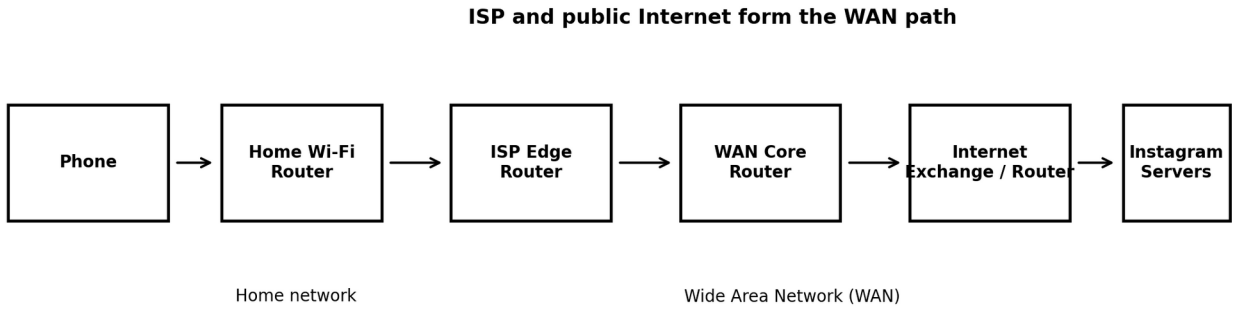
WIDE AREA NETWORK ASSIGNMENT

1. Draw a simple diagram showing how your phone connects to Instagram's servers from your home Wi-Fi, including the ISP and at least two WAN devices.



Explanation: The phone connects to the home Wi-Fi router, which sends the request to the Internet Service Provider (ISP). The request then travels through ISP and Internet routers across the WAN before reaching Instagram's servers. The response follows a network route back to the phone.

2. Create a table listing three types of WAN technologies and compare them based on speed, cost, and real-life use.

WAN technology	Speed	Cost	Real-life use
MPLS (Multiprotocol Label Switching)	Reliable, predictable performance; often supports high bandwidth.	High; usually requires a managed service contract.	A large company such as Zomato could connect offices, warehouses, and data centres over a managed private WAN.
Leased Line	Dedicated bandwidth with consistent performance and service guarantees.	High; generally more expensive than ordinary broadband.	A bank, large office, or IRCTC facility may use a leased line for dependable connectivity.
Broadband Internet	Can provide high download speeds, but performance may vary with congestion and service plan.	Low to medium; commonly affordable for homes and small offices.	Home Wi-Fi users streaming YouTube or Spotify, and small business offices.

Comparison: MPLS and leased lines are generally chosen when an organization needs dependable connectivity between locations. Broadband is usually cheaper and widely available, making it suitable for homes and many small offices.

3. Explain how a VPN uses WANs to let you access apps like Spotify or YouTube securely from public Wi-Fi at a coffee shop.

Answer: A public Wi-Fi network connects the device to the Internet through the coffee shop's ISP and the wider Internet WAN. A VPN client establishes an encrypted tunnel from the device to a VPN server across this public network. Data travelling through the tunnel is encrypted between the device and VPN server, making it harder for people on the same Wi-Fi network to read or modify the traffic. The VPN server then forwards the request to services such as Spotify or YouTube and sends the response back through the tunnel.

Important point: A VPN protects traffic between the device and the VPN server, but it does not make a user completely anonymous or protect against phishing, malware, or unsafe downloads. HTTPS should still be used.

4. Imagine Flipkart wants to connect its warehouses in Ahmedabad, Mumbai, and Delhi. Suggest the best WAN technology and justify your answer with two reasons.

Recommendation: A managed MPLS WAN would be a suitable choice for connecting Flipkart's warehouses in Ahmedabad, Mumbai, and Delhi.

Reason 1: MPLS can provide reliable and predictable connectivity between multiple business locations, helping warehouses access shared inventory, order-processing, and logistics systems.

Reason 2: A managed MPLS service can support traffic prioritization and service-level agreements (SLAs), helping important business applications maintain consistent performance.

Additional note: The final choice would depend on budget, bandwidth requirements, availability, and the company's network design. A secure SD-WAN solution using business broadband or leased-line links could also be considered as an alternative.